

SWIFT CURRENT AP, SK, Canada

WMO: 711420

Lat: 50.290N

Lon: 107.690W

Elev: 2680

StdP: 13.33

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-16.8	-30.2	1.1	-22.3	-23.9	1.6	-16.5	36.1	24.0	32.2	19.3	12.4	270	0.778

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	88.2	63.7	84.6	62.8	80.8	61.6	67.3	80.8	65.3	78.8	63.5	76.6	12.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	95.1	72.3	60.7	87.4	70.1	58.8	81.6	68.3	33.4	80.6	31.7	79.0	30.2	76.4	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.7	26.6	23.9	DB	-28.7	94.3	5.0	3.4	-32.3	96.8	-35.2	98.8	-38.0	100.7	-41.7	103.2	(4)
(5)			WB	-28.9	71.7	4.9	2.4	-32.5	73.4	-35.4	74.8	-38.1	76.1	-41.7	77.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.7	12.9	15.0	26.1	40.2	51.2	59.3	65.6	64.8	55.3	41.4	26.9	15.5	(6)
(7)		DBStd	22.11	15.91	14.42	14.54	10.27	8.66	6.48	5.88	6.98	9.05	9.96	12.98	14.67	(7)
(8)		HDD50	5398	1152	981	740	316	92	7	0	1	46	297	697	1069	(8)
(9)		HDD65	9466	1617	1401	1204	743	432	191	66	91	310	733	1144	1534	(9)
(10)		CDD50	1624	0	0	1	23	131	287	483	460	206	30	3	0	(10)
(11)		CDD65	216	0	0	0	0	5	21	84	85	21	0	0	0	(11)
(12)		CDH74	2851	0	0	0	10	124	280	982	1114	331	10	0	0	(12)
(13)		CDH80	942	0	0	0	1	22	65	323	424	107	0	0	0	(13)
(14)	Wind	WSAvg	12.7	14.2	13.2	13.4	13.1	12.9	11.7	10.6	10.7	11.9	13.0	13.5	14.1	(14)
(15)	Precipitation	PrecAvg	15.3	0.9	0.7	0.8	1.1	1.9	2.8	1.9	1.7	1.4	0.7	0.6	0.9	(15)
(16)		PrecMax	22.6	2.0	1.6	2.8	3.5	5.2	8.7	4.3	5.3	3.9	2.7	1.8	2.2	(16)
(17)		PrecMin	10.4	0.1	0.2	0.1	0.2	0.3	0.8	0.3	0.2	0.1	0.0	0.1	0.4	(17)
(18)		PrecStd	3.1	0.5	0.4	0.7	0.7	1.2	1.7	1.1	1.2	1.1	0.6	0.4	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.5	45.5	60.2	74.0	82.7	86.7	91.9	93.9	88.6	75.1	61.4	47.0	(19)
(20)			MCWB	38.6	38.8	47.3	52.8	57.5	64.3	66.0	64.1	60.7	56.3	47.5	37.9	(20)
(21)		2%	DB	40.0	40.3	54.1	67.5	77.5	81.1	87.4	89.1	83.1	67.2	52.8	41.5	(21)
(22)			MCWB	35.5	35.5	43.4	49.9	56.8	62.1	65.8	62.9	58.8	51.7	43.3	35.8	(22)
(23)		5%	DB	37.3	37.0	48.6	62.2	72.7	76.7	83.6	85.0	77.0	61.3	47.6	38.1	(23)
(24)			MCWB	33.9	33.4	40.5	46.8	54.0	60.2	65.1	61.8	57.1	48.5	39.9	33.6	(24)
(25)		10%	DB	34.1	33.8	43.2	57.2	68.0	72.9	79.7	80.4	71.5	56.1	42.5	34.5	(25)
(26)			MCWB	31.4	31.1	37.3	44.7	52.1	58.5	63.6	61.0	54.8	46.1	36.8	31.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.8	39.5	48.0	54.0	61.0	68.3	72.1	67.8	63.0	56.8	48.4	39.7	(27)
(28)			MCDWB	43.1	44.5	58.6	70.8	76.2	79.8	83.9	84.3	82.0	72.5	59.6	45.2	(28)
(29)		2%	WB	35.8	35.9	44.3	50.8	57.6	64.3	68.7	65.6	60.3	52.7	44.0	36.1	(29)
(30)			MCDWB	39.4	39.4	52.9	64.9	74.1	75.8	81.2	80.8	77.4	64.2	52.2	41.0	(30)
(31)		5%	WB	34.0	33.6	40.8	48.0	55.5	61.9	66.4	64.0	58.1	49.5	40.3	33.7	(31)
(32)			MCDWB	37.1	36.8	48.2	60.7	69.7	73.1	79.3	79.2	74.3	59.7	46.7	37.9	(32)
(33)		10%	WB	31.3	31.1	37.5	45.4	53.2	59.8	64.6	62.4	55.8	46.7	37.1	31.0	(33)
(34)			MCDWB	34.1	33.7	43.0	55.7	65.5	70.1	77.1	76.8	70.0	55.0	42.0	34.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.0	16.3	17.0	21.0	22.5	20.0	23.0	24.1	22.5	19.5	16.0	15.1	(35)
(36)			MCDBR	17.1	17.5	23.4	28.9	29.4	26.7	28.7	31.6	31.4	27.6	22.0	17.2	(36)
(37)			MCWBR	14.1	13.8	14.7	15.4	14.0	12.8	12.7	12.3	13.9	15.1	14.3	13.1	(37)
(38)		5% WB	MCDWB	16.8	17.2	22.3	27.0	26.5	23.5	25.2	26.2	27.9	25.5	21.5	17.4	(38)
(39)			MCWBR	14.2	13.9	14.5	15.1	13.2	12.3	12.8	11.4	13.0	14.6	14.4	13.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.245	0.246	0.276	0.343	0.364	0.357	0.364	0.383	0.333	0.286	0.256	0.234	(40)
(41)		taud		2.417	2.484	2.503	2.322	2.309	2.386	2.381	2.291	2.407	2.530	2.500	2.445	(41)
(42)		Ebn at Noon		262	293	300	285	283	285	281	269	273	272	256	252	(42)
(43)		Edh at Noon		19	23	27	36	39	36	36	37	30	22	17	16	(43)
(44)	All-Sky Solar Radiation	RadAvg		385	704	1126	1536	1845	1951	2118	1708	1249	766	418	295	(44)
(45)		RadStd		41	55	82	110	132	133	130	129	113	88	47	29	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page